Query Analysis

Table 1: Display Courses Table

SQL Query

SELECT * FROM courses;

Result

Displays all courses available in the database along with their unique Course IDs.

Explanation

This query retrieves all records from the Courses table. It is used to verify that the course details have been inserted successfully into the database.

	id [PK] integer	name character varying (50)
1	1	Mathematics
2	2	Science
3	3	English
4	4	Computer
5	5	Physics

Table 2: Display Enrollments Table

SQL Query

SELECT * FROM enrollments;

Result

Displays all student enrollment records, including the Student ID, Course ID, and the grade obtained in each course.

Explanation

This query retrieves every record from the Enrollments table. It verifies that students have been correctly linked to their respective courses and that their grades have been stored successfully.

	student_id integer	course_id integer	grade integer
1	1	1	90
2	1	2	85
3	1	3	88
4	2	1	76
5	2	2	80
6	2	4	91
7	3	1	45
8	3	3	60
9	3	5	70
10	4	2	35
11	4	3	55
12	4	4	65
13	5	1	92
14	5	4	89
15	5	5	94
16	6	2	41
17	6	3	38
18	6	5	52
19	7	1	83
20	7	2	79
21	7	4	86
22	8	3	90
23	8	4	95

Query 1: List All Students Enrolled in Each Course

SQL Query

```
SELECT
    s.StudentName,
    c.name AS CourseName,
    e.grade
FROM enrollments e
JOIN students s
ON e.student_id = s.StudentsID
JOIN courses c
ON e.course_id = c.id
ORDER BY c.name, s.StudentName;
```

Result

Displays the name of each student, the course they are enrolled in, and their corresponding grade.

Explanation

This query uses INNER JOIN to combine the Students, Courses, and Enrollments tables. It retrieves student names, course names, and grades and sorts the results alphabetically by course and student name.

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	studentname character varying (100)	coursename character varying (50)	grade integer
1	Aditya Singh	Computer	86
2	Ananya Joshi	Computer	65
3	Diya Shah	Computer	91
4	Krish Desai	Computer	89
5	Meera Nair	Computer	95
6	Sneha Kapoor	Computer	75
7	Aarav Patel	English	88
8	Ananya Joshi	English	55
9	Meera Nair	English	90
10	Rahul Verma	English	40
11	Riya Patel	English	38
12	Vivaan Mehta	English	60
13	Aarav Patel	Mathematics	90
14	Aditya Singh	Mathematics	83
15	Diya Shah	Mathematics	76
16	Krish Desai	Mathematics	92
17	Rahul Verma	Mathematics	28
18	Sneha Kapoor	Mathematics	69
19	Vivaan Mehta	Mathematics	45
20	Krish Desai	Physics	94
21	Meera Nair	Physics	88
22	Riya Patel	Physics	52
23	Sneha Kapoor	Physics	81

Query 2: Average Grade Per Course

SQL Query

```sql
SELECT
   c.name AS CourseName,
   ROUND(AVG(e.grade),2) AS Average_Grade
FROM enrollments e
JOIN courses c
ON e.course_id = c.id
GROUP BY c.name
ORDER BY Average_Grade DESC;
```

Result

Displays the average grade obtained by students in each course.

Explanation

This query calculates the average grade for every course using the AVG() aggregate function. The GROUP BY clause groups the records according to each course, and ROUND() formats the average grade to two decimal places.

	coursename character varying (50)	avg_grade numeric
1	Computer	83.50
2	Physics	77.00
3	Mathematics	69.00
4	English	61.83
5	Science	58.83

Query 3: Top 3 Students Overall

SQL Query

```sql
SELECT
   s.StudentName,
   ROUND(AVG(e.grade),2) AS Average_Grade
FROM enrollments e
```

JOIN students s

ON e.student_id = s.StudentsID

GROUP BY s.StudentName

ORDER BY Average_Grade DESC

LIMIT 3;

Result

Displays the top three students based on their average grades across all enrolled courses.

Explanation

This query calculates each student's average grade using the AVG() function. The records are sorted in descending order, and the LIMIT clause returns only the top three students.

	studentname character varying (100)	averagegrade numeric
1	Krish Desai	91.6666666666666667
2	Meera Nair	91.0000000000000000
3	Aarav Patel	87.6666666666666667

Query 4: Count Students Who Failed

SQL Query

SELECT

COUNT(DISTINCT student_id) AS Failed_Students

FROM enrollments

WHERE grade < 40;

Result

Displays the total number of students who scored below 40 in at least one course.

Explanation

This query uses the COUNT(DISTINCT) function to count unique students who failed at least one course. The WHERE clause filters only those records where the grade is less than 40.

	failedstudents bigint
1	3